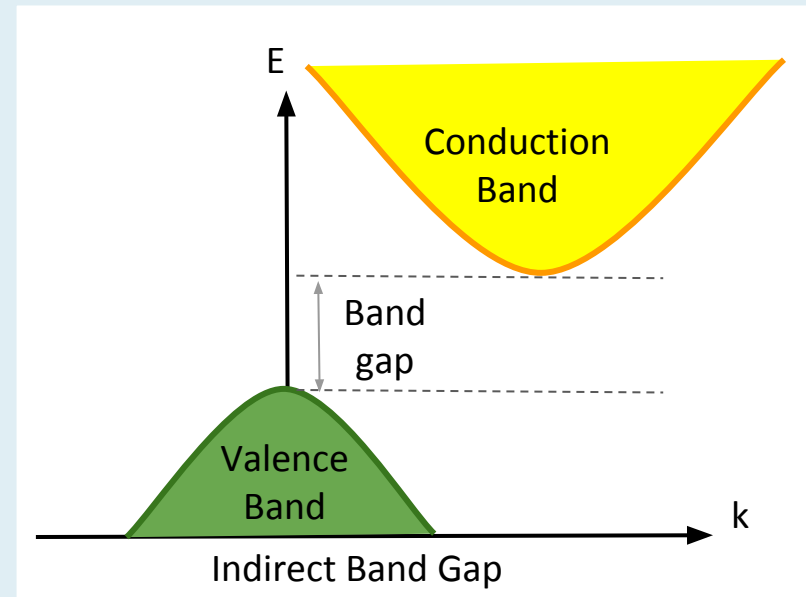
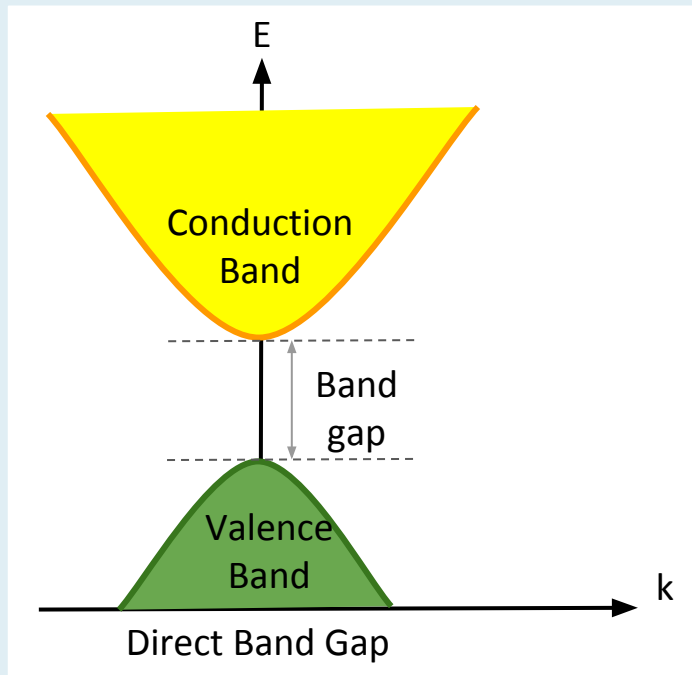


SEMICONDUCTOR

Direct and Indirect Bandgap

1 time for 3 Marks

- **Indirect band gap semiconductors:** semiconductors in which the electron from the conduction band comes to the hole in the valence band via trapped energy state.
- **Direct band gap semiconductors:** semiconductors in which the electrons directly recombine with the hole in the valence band.



Terms involved in semiconductors

3 times for 3 Marks

- **Recombination:** it is the process when electron in the conduction band releases energy and combines with its respective hole in the valence band.
- **Drift velocity:** it is the velocity of electron in the steady state in an applied electric field.
- **Mobility of electrons:** it is the drift velocity per unit electric field.
- **Diffusion current:** non uniform concentration of charge carriers produce diffusion current.
- **Current density:** it is the current per unit area of cross section
- **Electrical field:** it is the potential drop per unit length of the conductor
- **Conductivity:** it is the physical property that represents the conduction ability of material
- **Resistivity:** it is the inverse of conductivity.

Concept Of Holes

2 times for 3-4 Marks

- A completely filled energy band does not conduct electricity.
- However if some electrons are excited and shifted to the conduction band, the remaining electrons contribute in conduction.
- The electrons which move to the conduction band leave *positively charged vacant spaces*(quasiparticles) in the valence band: holes.
- They move in the direction opposite to that of electrons when subjected to an electrical field.
- Current in a semiconductor = motion of *electrons in conduction* band + motion of *holes in valence* band.
- The motion of electron in the field is given by $F = m_e a$ or (effective mass) m_e^* .
- Energy levels in the conduction band are vacant, whereas energy levels in the valence band are filled. Thus electrons move freely but holes cannot. Hence *effective mass of holes is greater than the effective mass of electrons*.

Fermi Level And Fermi-dirac Statistics

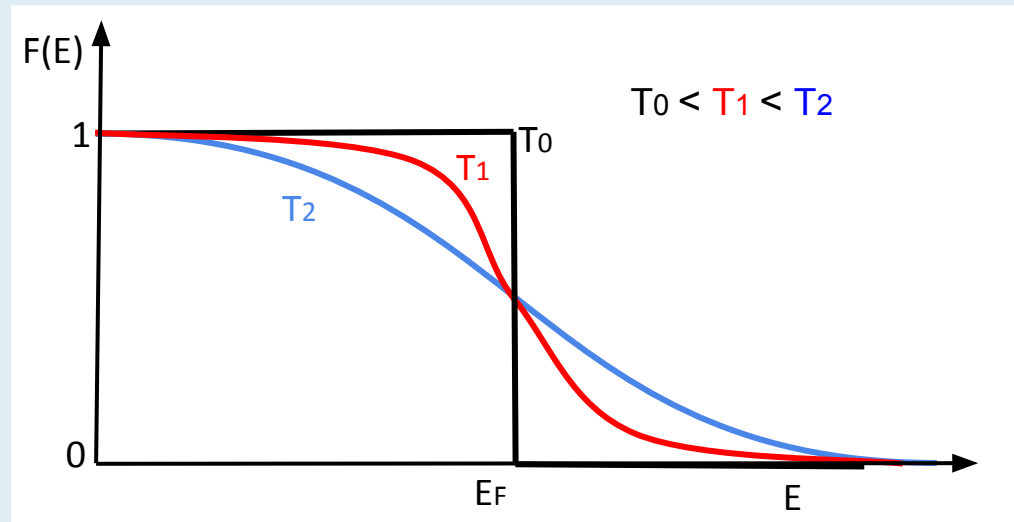
5 times for 3-5 Marks

- The highest occupied energy level at 0K is called the Fermi energy and the level is called Fermi level E_F .
- According to Fermi-Dirac statistics, we use a function $f(E)$ which determines the carrier occupancy of the energy.
- $f(E) = 1 / (1 + \exp[(E - E_F)/KT])$
- $f(E)$ = probability that a particular energy level is occupied by an electron.
- At normal temperatures a significant amount of electrons get excited to the conduction band and leave exact number of holes in valence band.
- Therefore $f(E)$ has a non zero probability above Fermi level and reduces by the same amount below E_F .

Fermi Level In Semiconductors

5 times for 3-5 Marks

- **At $T = 0\text{K}$:- case 1 : $E < E_F$ $f(E) = 1$** , which means that all the levels below E_F are occupied by electrons.
- **At $T = 0\text{K}$:- case 2 : $E > E_F$ $f(E) = 0$** , which means all the levels above E_F are vacant.
- **At $T = 0\text{K}$:- case 3 : $E = E_F$ $f(E)$ cannot be determined.**
- **At $T > 0\text{K}$:- case 1 : $E = E_F$ $f(E) = \frac{1}{2}$** , few levels just below E_F will be vacant.



Intrinsic Semiconductors

9 times for 3-8 Marks

- Pure semiconductors are called intrinsic semiconductors.
- The conduction in intrinsic semiconductors is due to the thermal excitation of electrons and holes.
- Conduction is small as compared to extrinsic semiconductors.
- Fermi level in intrinsic semiconductors lies exactly at the centre of the forbidden energy gap.
- This is because for intrinsic semiconductors number of electrons in conduction band is equal to the number of holes in the valence band.
- $n_e = n_h = n_i$
- Here n_i is intrinsic carrier concentration.

Extrinsic Semiconductors

- When some impurity is added to a pure semiconductor to increase its conductivity, it is called extrinsic semiconductor.
- These impurities are generally group III or group V elements from the periodic table.
- If a pentavalent impurity is added then we call it n-type semiconductor.
- If a trivalent impurity is added then we call it p-type semiconductor.
- These impurities are called dopants and the process of adding impurities to a pure semiconductor is called doping.
- The impurities are of substitutional type. Therefore the original crystal structure does not get distorted.

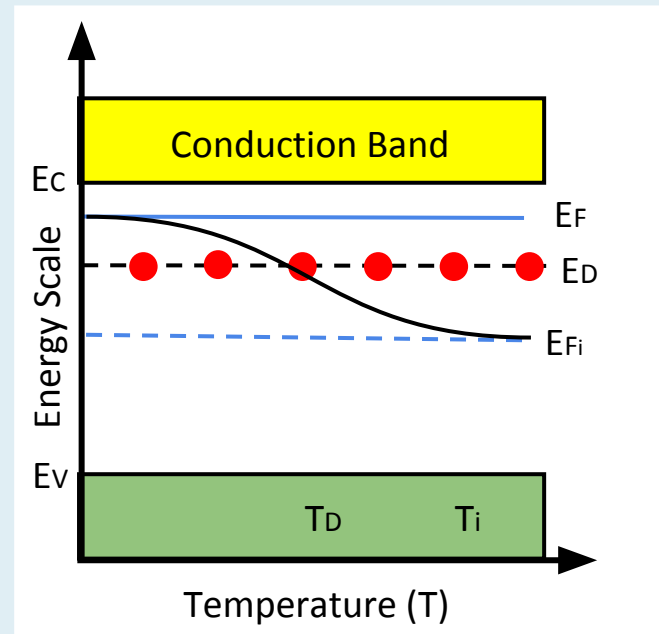
Difference b/w Intrinsic And Extrinsic Semiconductors

INTRINSIC SEMICONDUCTOR	EXTRINSIC SEMICONDUCTOR
It is a pure crystal.	It is a single crystal with impurities.
The charge carriers are majorly electrons of the parent atom.	The charge carriers are electrons or holes of the impurity atom.
Number of electrons and holes are equal	In n-type the electrons are major charge carriers and in p-type holes are major charge carriers
More energy is required for electrons to cross E_g	Less energy is required for electrons to cross E_g

Effect Of Temperature On N-type Semiconductors

- **At low temperature** : only few donor atoms get ionised and hence the Fermi level lies midway between the bottom of conduction band and donor level.
- **At moderate temperature** : all donor atoms get ionised , so the Fermi level moves towards the centre of forbidden gap.
- **At high temperature** : the concentration of electrons transferred from valence to conduction band is higher than that from the donor atoms. Hence Fermi level shifts to the middle of the forbidden gap.

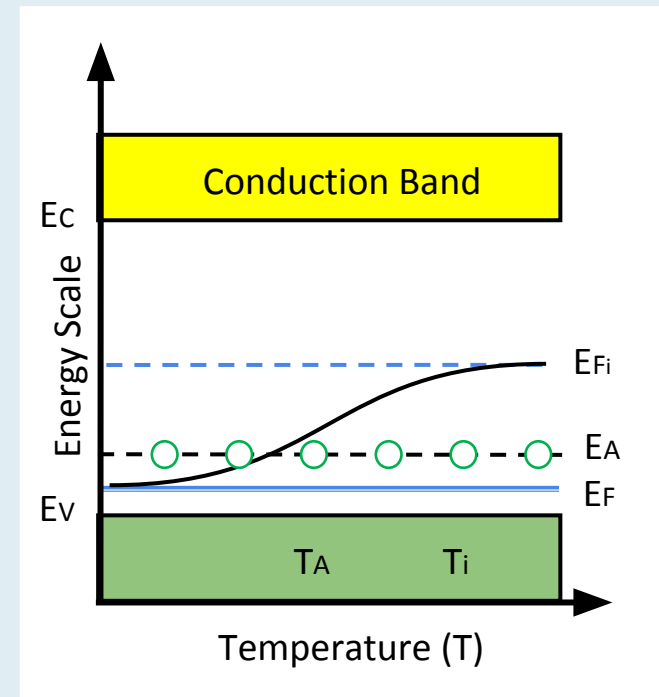
3 times for 3-5 Marks



Effect Of Temperature On P-type Semiconductors

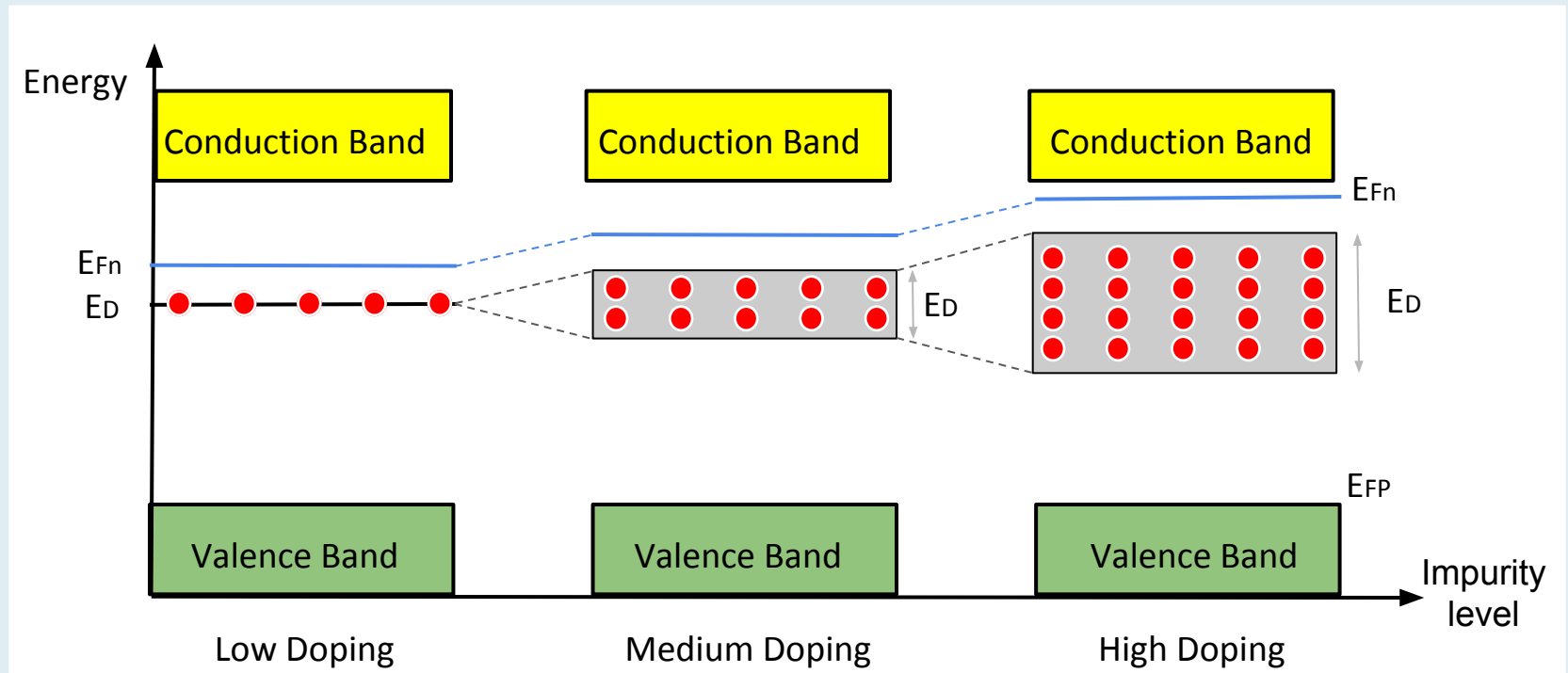
- **At low temperature** : only few acceptor levels are occupied , so the Fermi level lies in the middle of the top of the valence band and acceptor level.
- **At moderate temperature** : all acceptor levels are filled and hence the Fermi level moves towards the middle of the forbidden gap.
- **At high temperature** : the contribution of conduction band for formation of holes in the valence band is more compared to acceptor impurity. Therefore the Fermi level approaches the middle of the energy gap.

3 times for 3-5 Marks



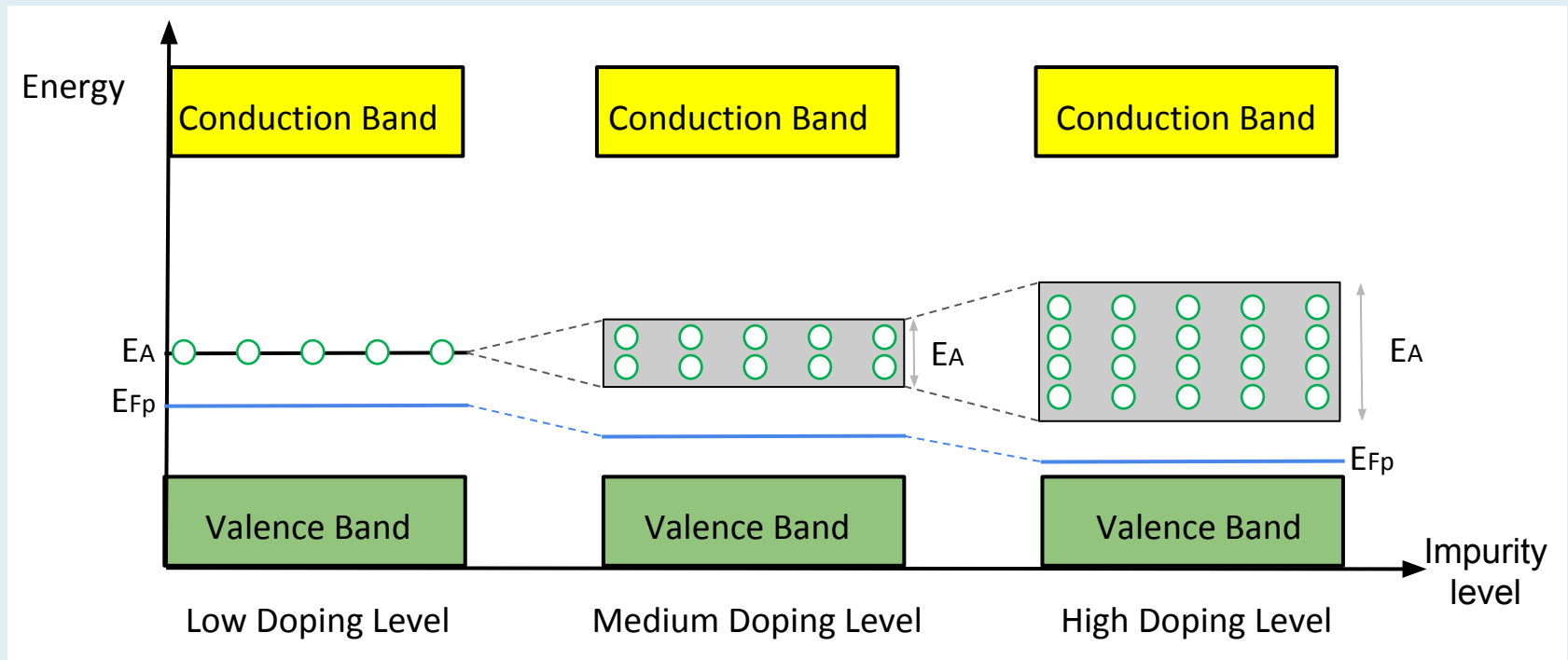
Effect Of Impurity Concentration On Fermi Level of N-type Semiconductor

2 times for 3-5 Marks



Effect Of Impurity Concentration On Fermi Level of P-type Semiconductor

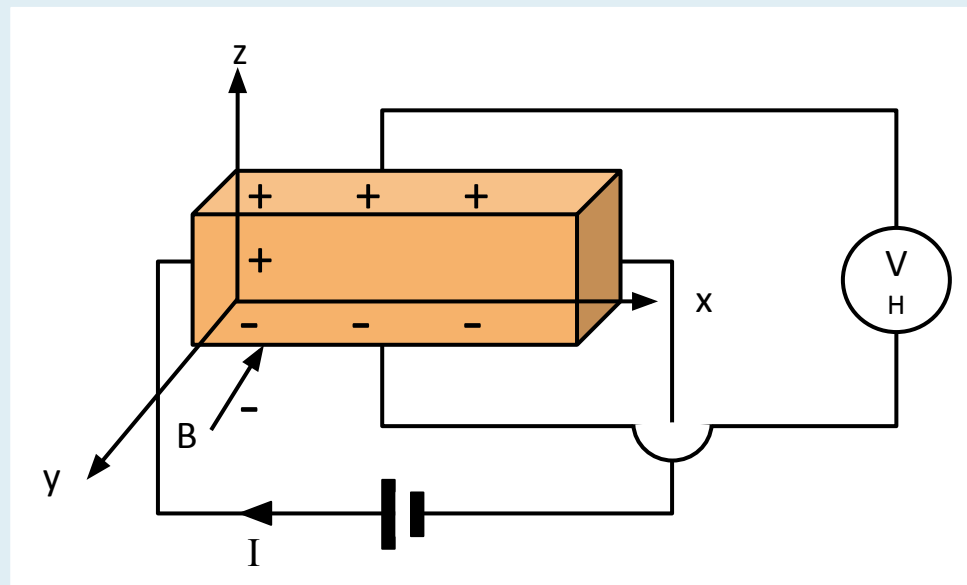
2 times for 3-5 Marks



Hall Effect

9 times for 3-8 Marks

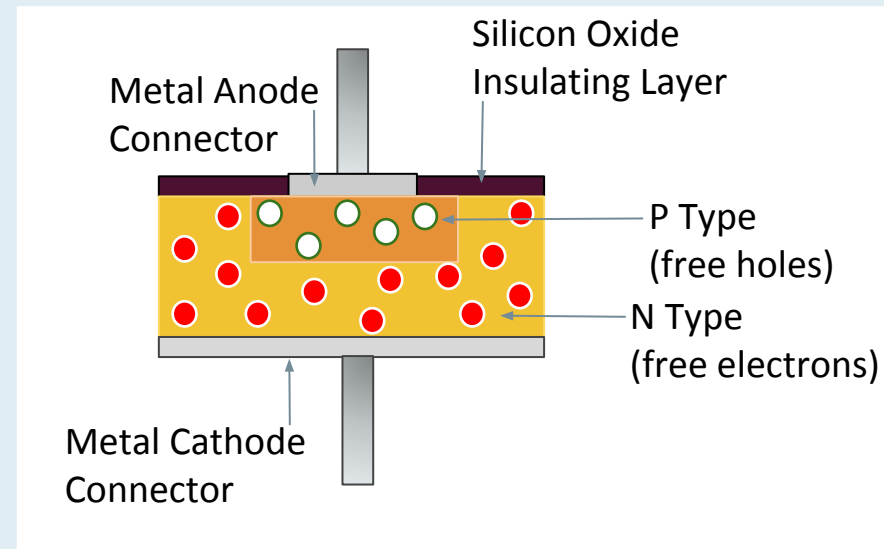
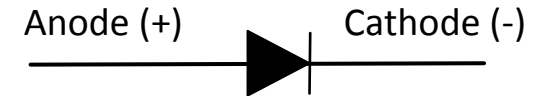
- If a metal or semiconductor carrying a current I is placed in a transverse magnetic field B , an electric emf E is induced in the direction perpendicular to both I and B .
- This phenomenon is known as Hall effect and the voltage induced is Hall voltage (V_H).
- **Applications:**
 - To determine concentration of charge carriers.
 - To determine mobility of charge carrier
 - To determine nature of semiconductor.
 - Used in magnetic field meter.

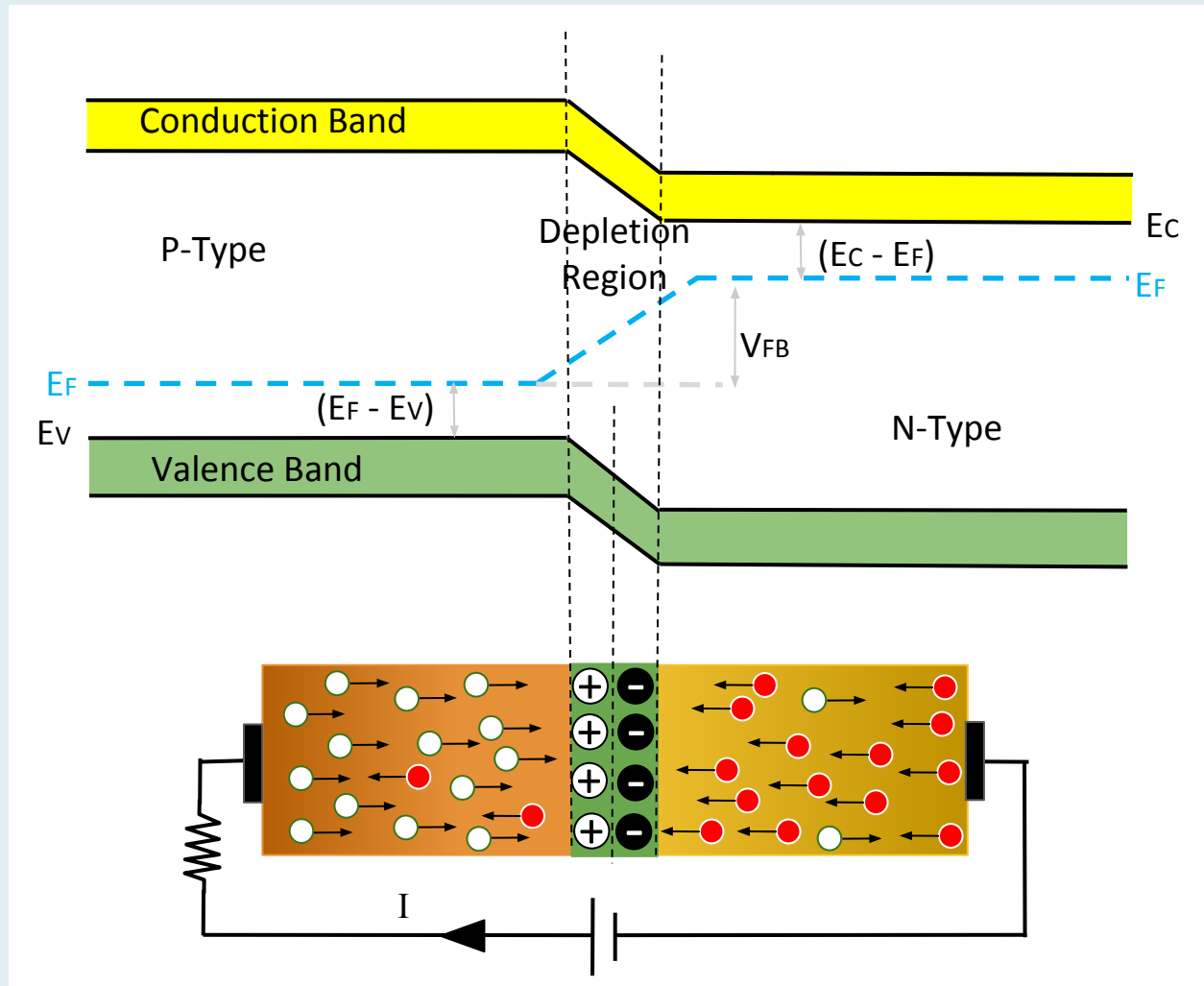


P-N Junction Diode: Forward Bias

6 times for 3-5 Marks

- The PN junction is forward biased when the P-side is connected to positive of the battery and the N-side is connected to the negative of the battery.
- Electrons from the battery pour into the N-side raising the fermi level there wherein the electrons are drained from the P-side dropping the fermi level on this side.
- This is indicated by the blue dotted line depicting fermi level for forward bias PN junction in figure.



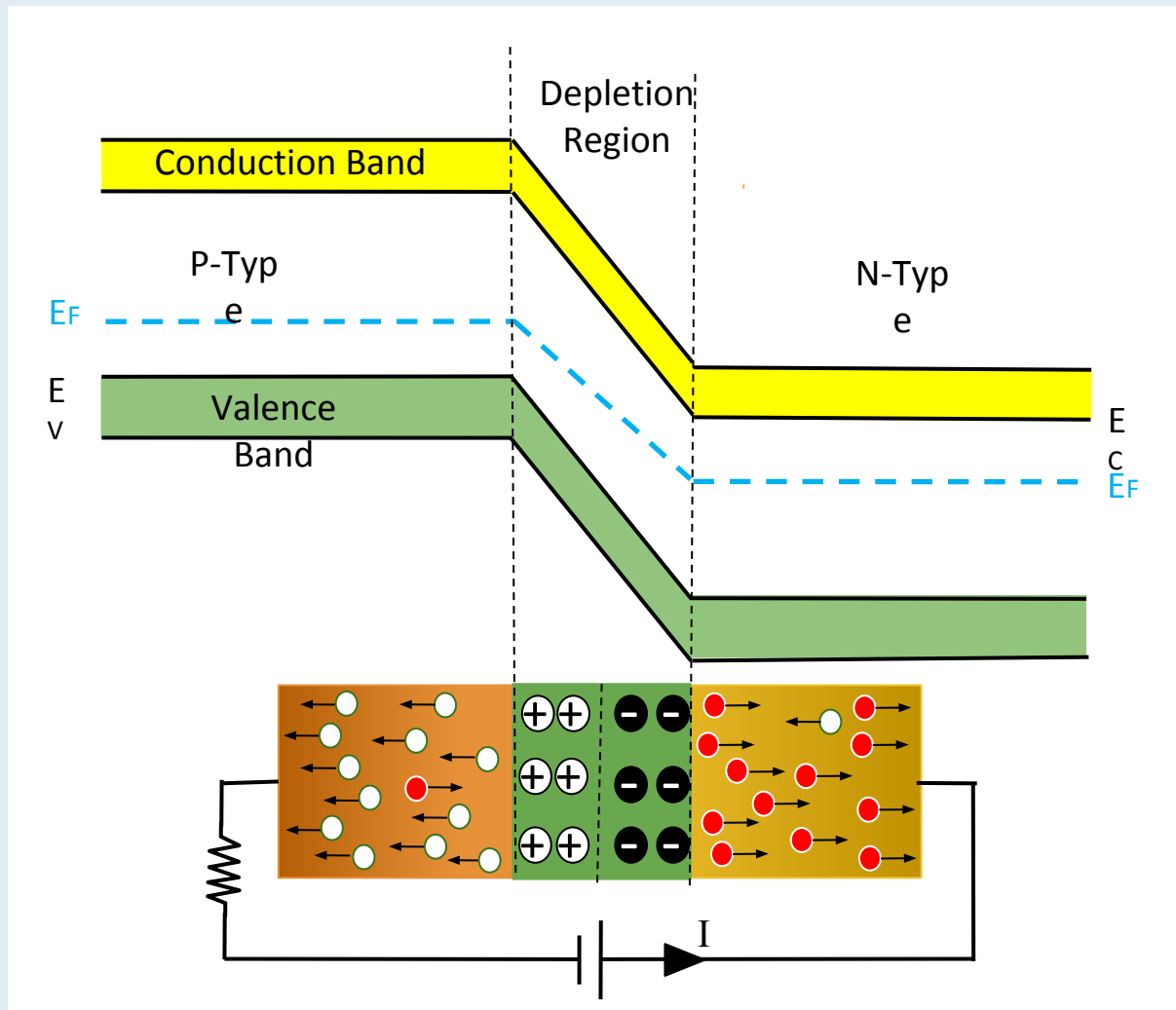


PN JUNCTION FORWARD BIAS

P-N Junction Diode: Reverse Bias

6 times for 3-5 Marks

- The PN junction is reverse biased when the N-side is connected to positive of the battery and the P-side is connected to the negative of the battery.
- Electrons from the battery pour into the P-side raising the fermi level there wherein the electrons are drained from the N-side dropping the fermi level on this side.
- This is indicated by the blue dotted line depicting fermi level for Reverse bias PN junction in figure.

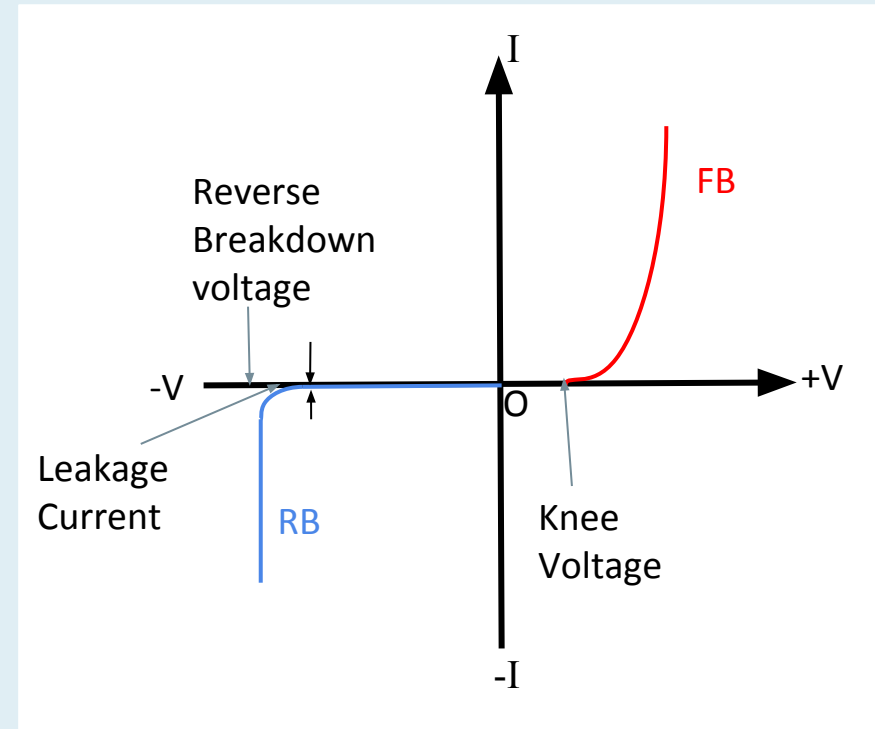


PN JUNCTION REVERSE BIAS

P-N Junction Diode: Unbiased

6 times for 3-5 Marks

- As the PN junction comes to existence, the electrons from N-side diffuse to P-side and recombine with holes.
- Electron hole pair recombination at the junction creates a region devoid of charge carriers left with immobile ions only is called the depletion region or the space charge region.
- The immobile ions in the depletion region create an electric field which further opposes the diffusion of electrons to P-side. Thus, halting the electron hole pair recombination and thereby restricting the growth of depletion region width.
- The width of the depletion region depends on the doping concentration. Higher the doping concentration, stronger is the field created in the space charge regions and hence thinner is the depletion region.



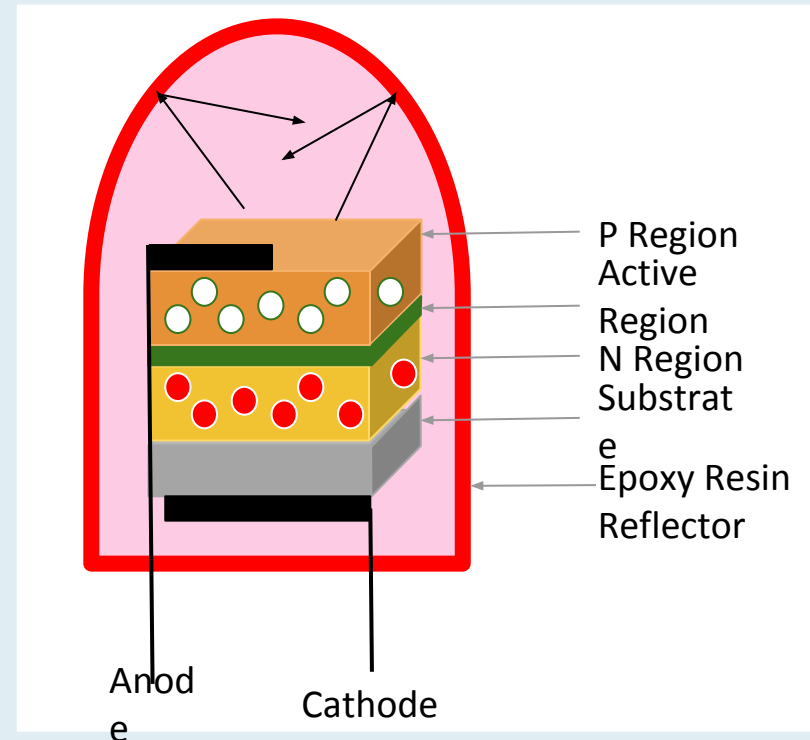
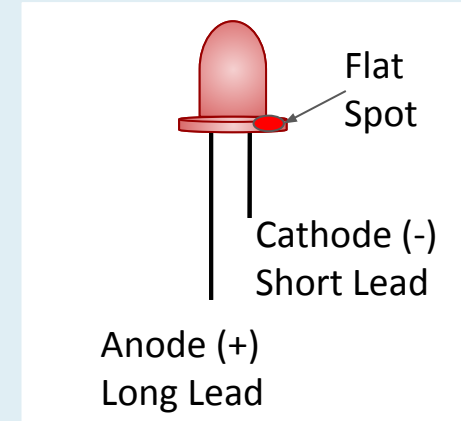
LED

- LED or **L**ight **E**mitting **D**iode is a two terminal device which emits light when supplied with electric potential.
- Functions only in forward bias.

CONSTRUCTION:

- Two semiconductor layers stacked on a substrate.
- Light emerges from the active region when recombination takes place.
- The p-region is kept above the n-region.
- Structure is placed inside a small reflective hemispherical cup made from a transparent plastic of epoxy resin. It helps focus all light.
- The cathode is sometimes coated with some reflective material (like gold) to reflect any light which reached the bottom by chance, back to the top.

2 times for 5 Marks



LED

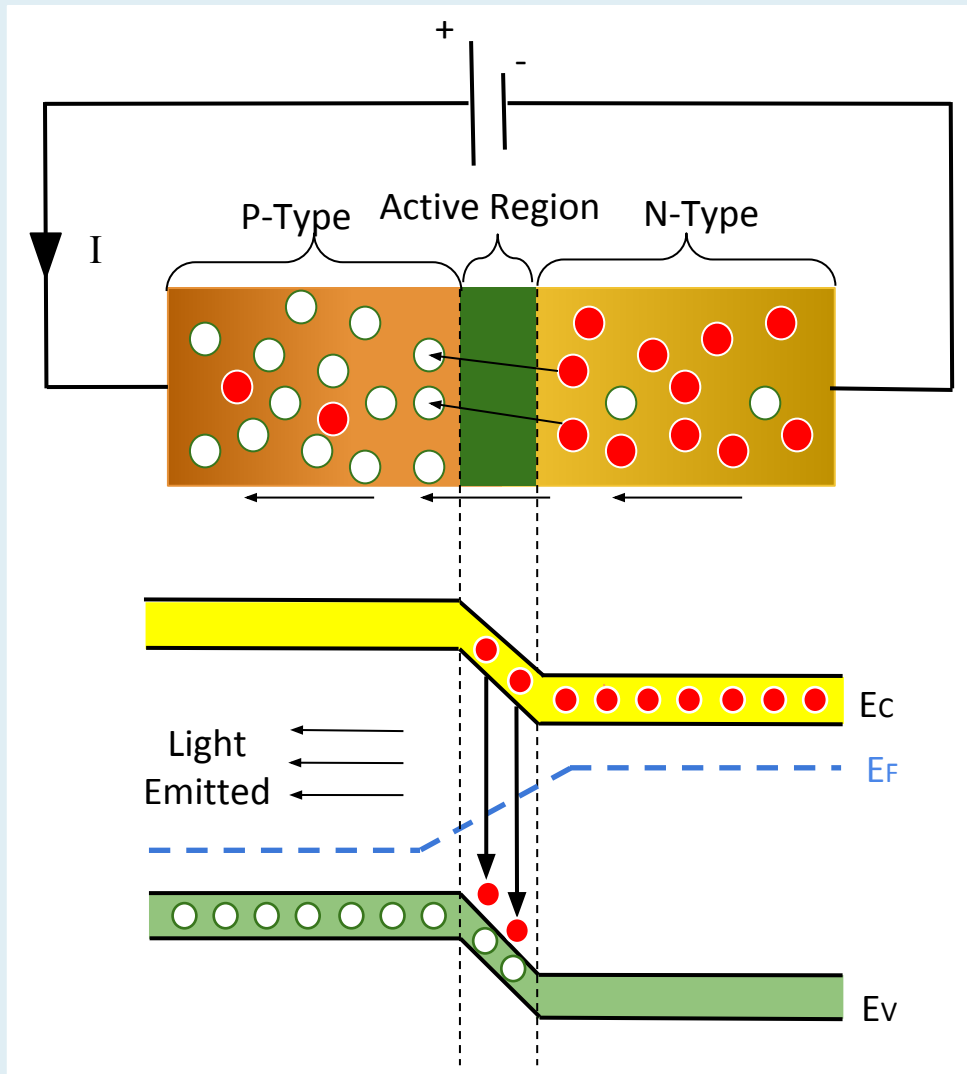
2 times for 5 Marks

WORKING:

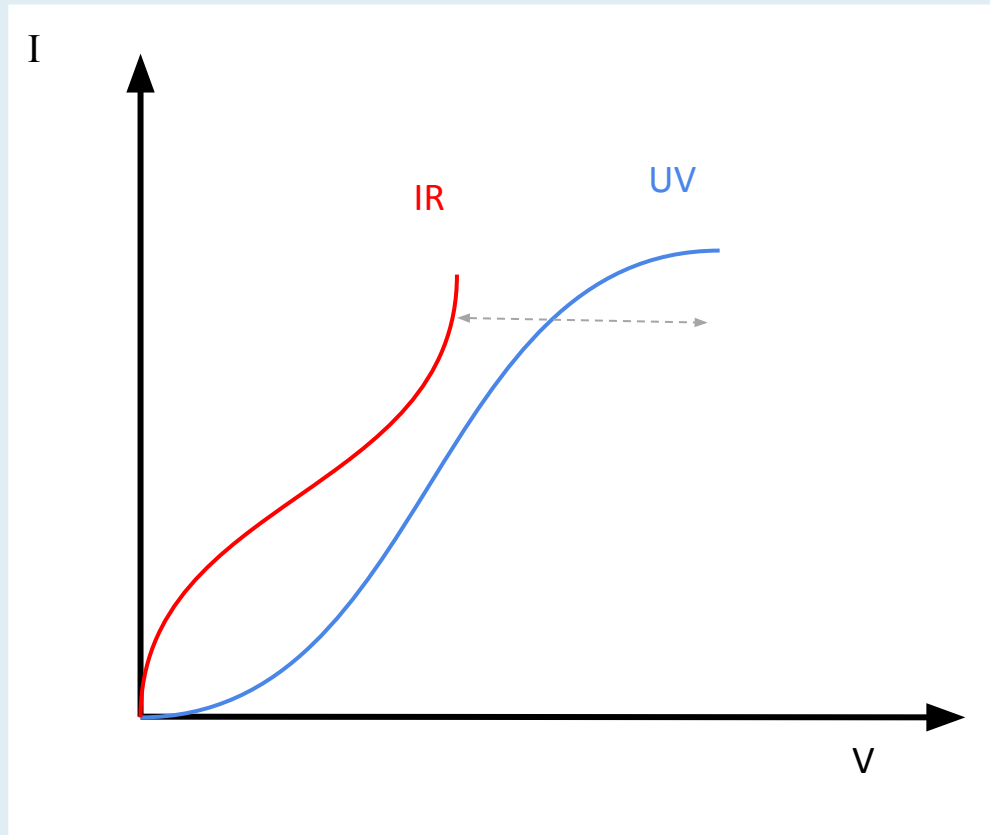
- **Principle: Electroluminescence**
- Forward bias is applied to the LED.
- P and N regions become aligned to allow recombination.
- Electrons are repulsed by the battery, which causes them to move from the conduction band into the valence band to recombine with the holes.
- Conduction band has a higher energy than valence band, the electrons move from a higher energy level, to a lower energy level.
- Electrons thus emit energy in the form of light.

IV CHARECTERISTICS:

As the wavelength of light decreases, the values of forward voltage increase for the same amount of current.

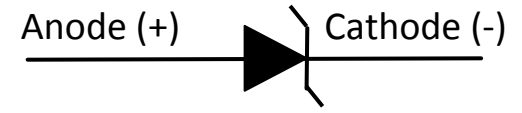


WORKING OF LED



I-V CHARACTERISTICS OF LED

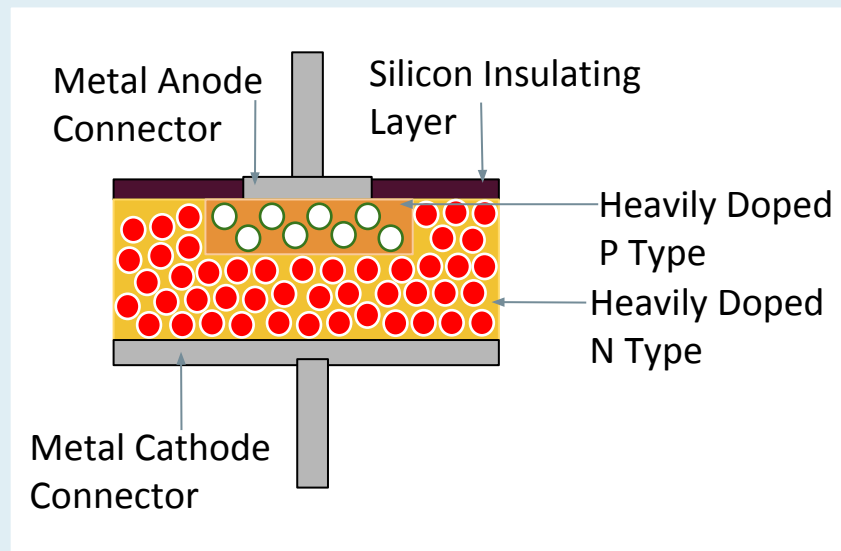
Zener Diode



- Diode which works in the breakdown region of the reverse bias mode.
- Used for voltage regulation.

CONSTRUCTION:

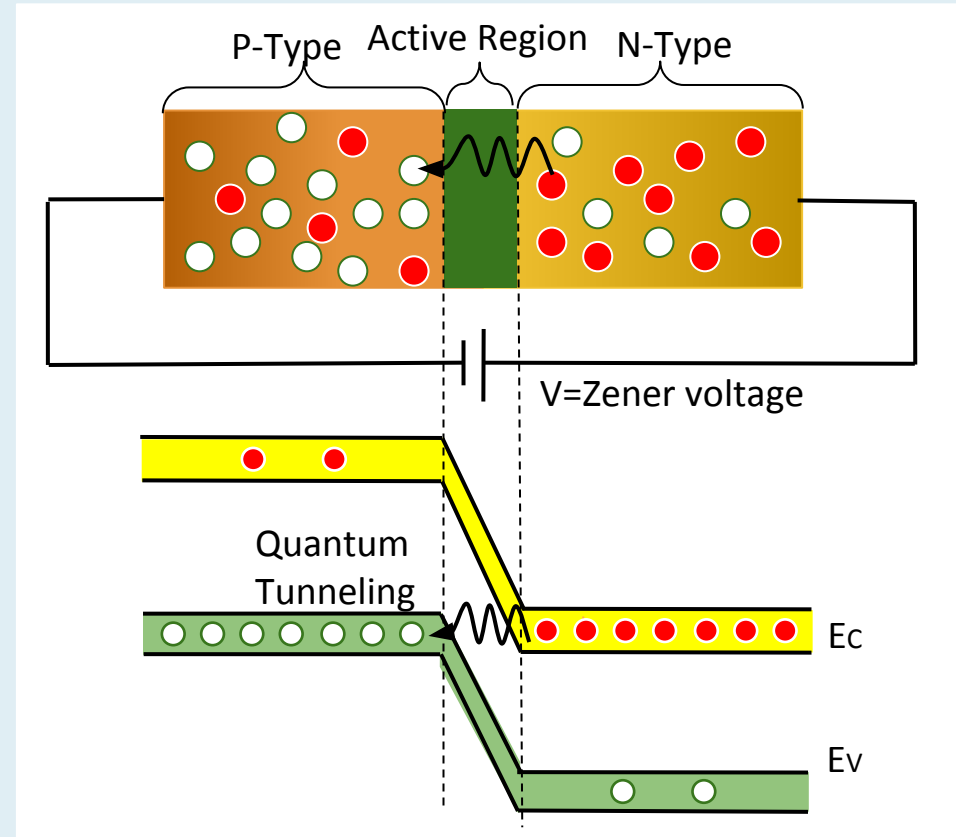
- A Zener diode is, in essence, a silicon semiconductor which has been metalized to facilitate connections to the anode and cathode.
- It consists of a heavily doped N-type and P-type substrate, such that the junction between the two is extremely thin.
- It is insulated with a layer of silicon dioxide to stop the contamination of the junction.

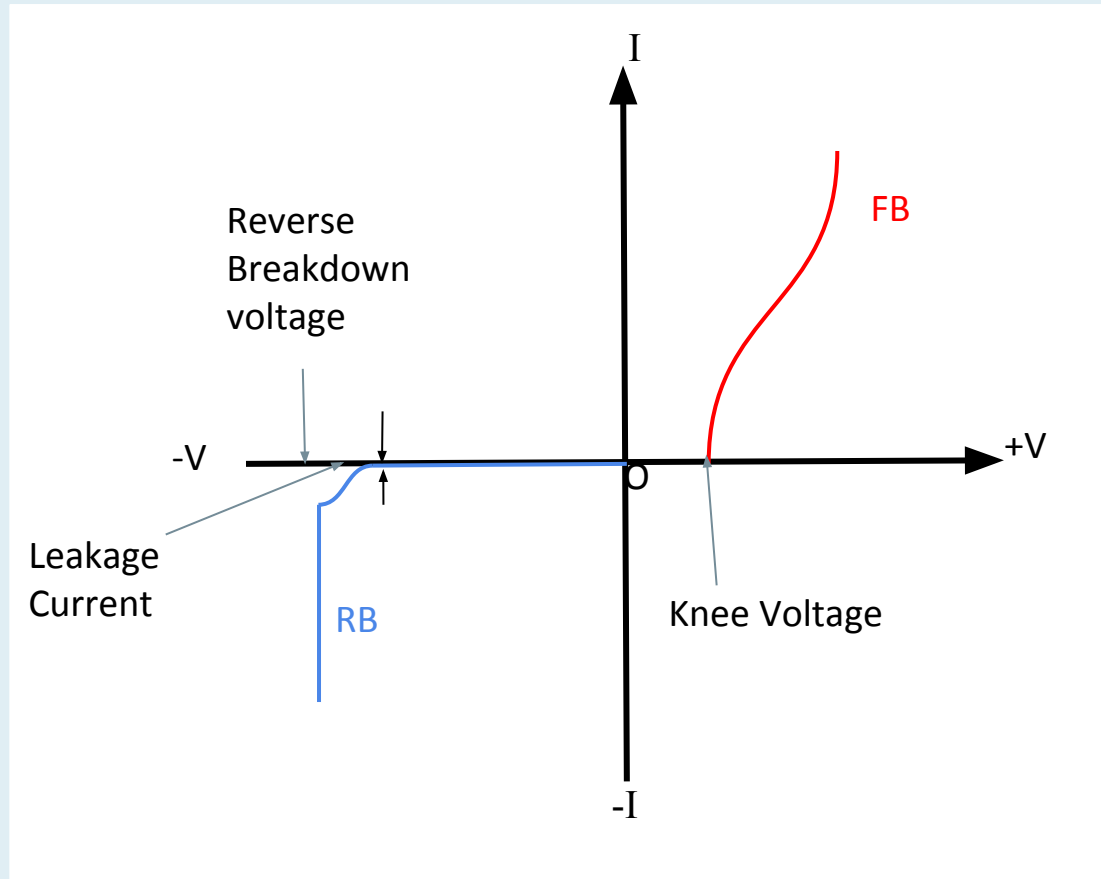


Zener Diode

WORKING:

- Due to heavy doping the depletion region becomes thin and breakdown occurs at low voltage.
- **Avalanche effect:** Takes place when the depletion region becomes wider due to reverse biased mode.
- At zener voltage, energy level of the valence band of the P-type conductor is equal to the energy level of the conduction band of the N-type conductor.
- Because of the narrow width, they “tunnel” through over to the valence band of the P-type conductor. This is the quantum tunnelling effect.
- A sharp rise in current is obtained. This is Zener Breakdown.



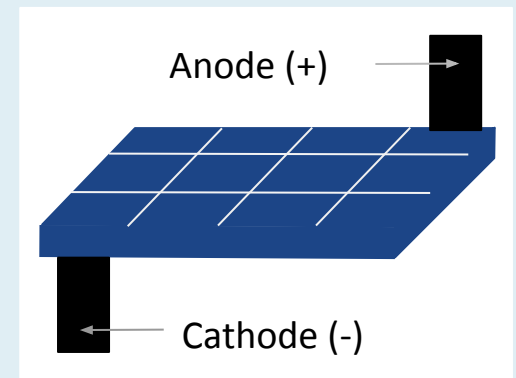
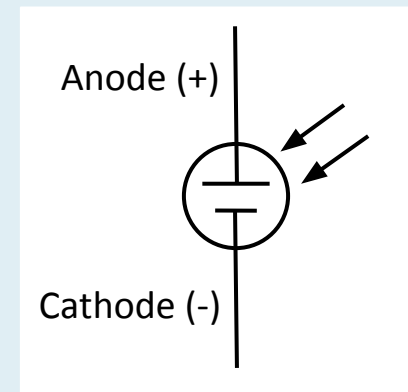
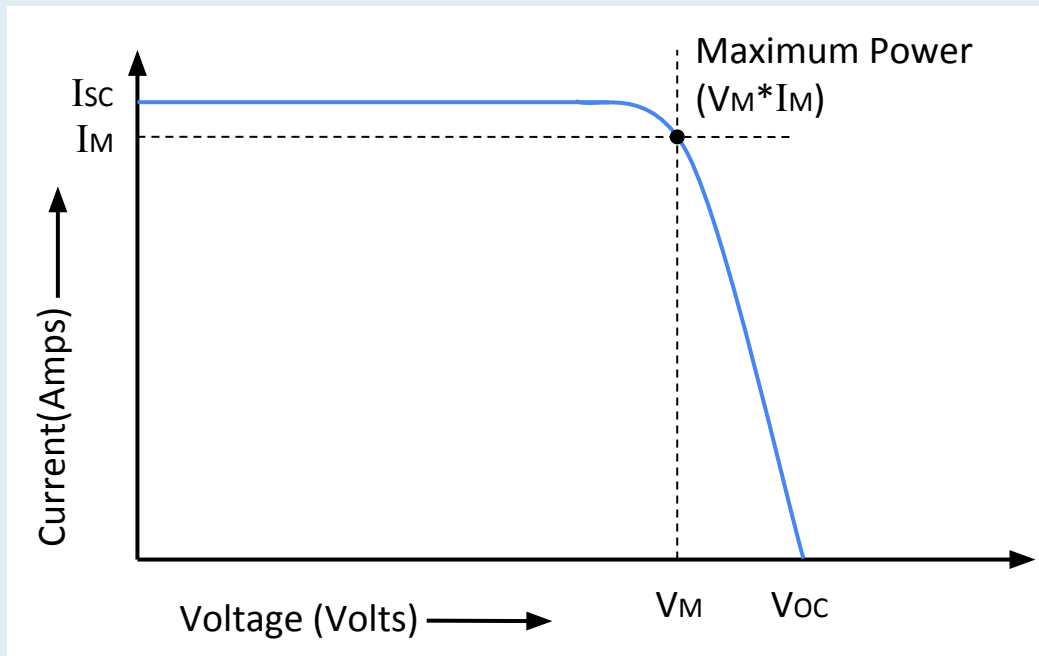


I-V CHARACTERISTICS OF ZENER DIODE

Solar Cell

9 times for 5 Marks

- A solar cell is a device which works on the principle of photovoltaic effect i.e. it converts incoming solar energy into electrical energy.

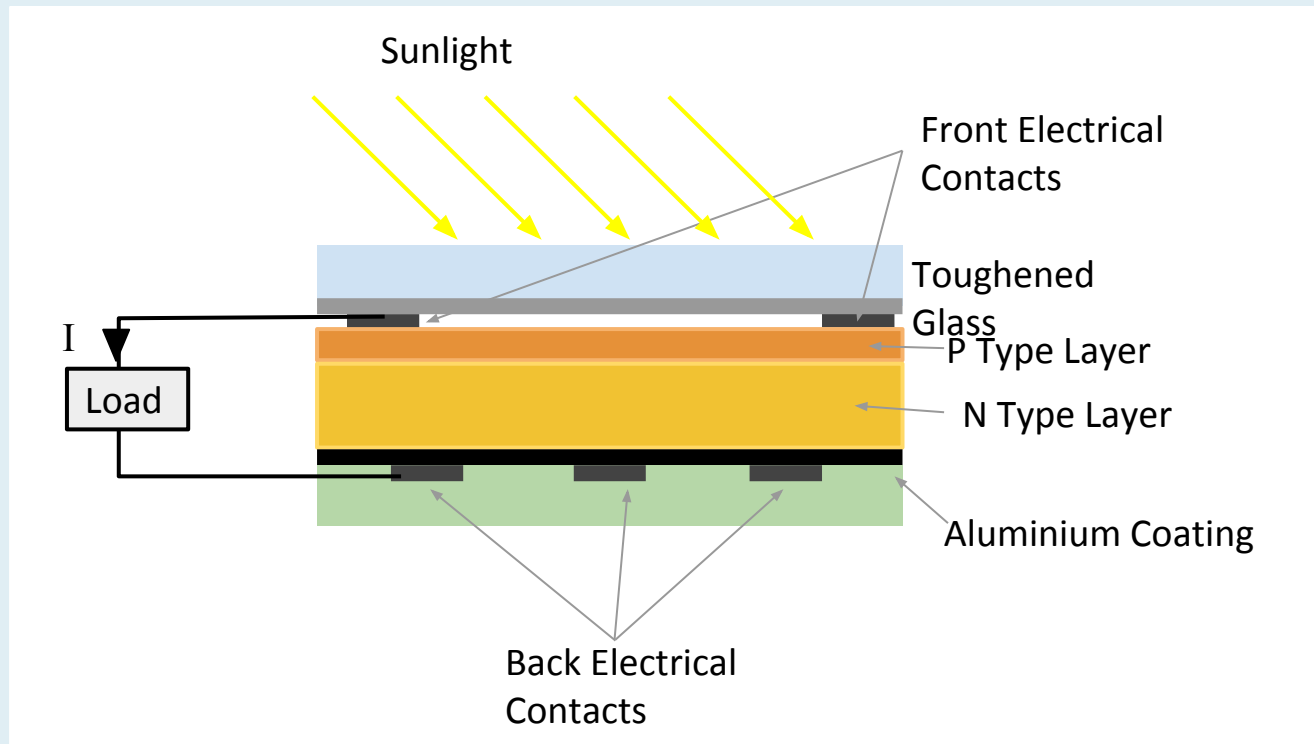


Solar Cell

9 times for 5 Marks

CONSTRUCTION :

- A solar cell is basically a p-n junction connected to an external circuit
- The p side of the cell is made thin as compared to the n side.
- This is to allow the solar energy ,incident on the p side, to reach the junction.

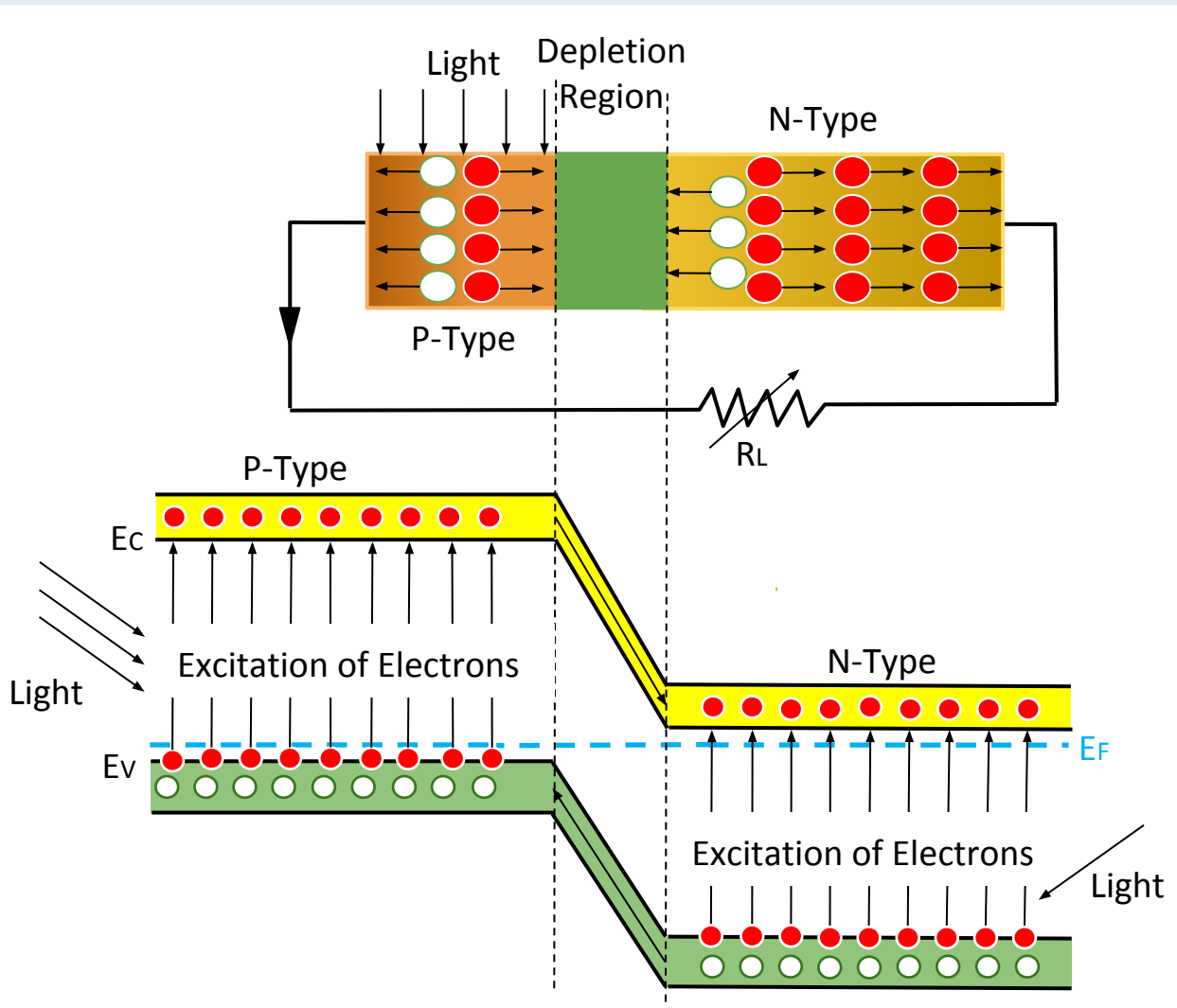


Solar Cell

9 times for 5 Marks

WORKING :

- In the equilibrium state the Fermi level lies near the conduction band of the n region and the valence band of the p region.
- When energy in the form of solar radiation falls on the p-n junction, the electrons from the valence band get excited and reach the conduction band as shown.
- Subsequently, holes are formed in the valence band.
- The flow of these charge carriers results in the formation of current in solar cells.



WORKING OF SOLAR CELL